



Generic vibration criteria for hand-held precision instruments

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ARTICLE INFO

Keywords:

Hand-arm vibration
Hand-held precision instrument
Vibration criteria
Physiological tremor
Inertial measurement
Power spectral density
Disturbance modeling
Optomechatronic system design

ABSTRACT

This paper presents a measurement-based framework for deriving generic hand-arm vibration criteria (HAVC) for the dynamic evaluation, modeling, and simulation of hand-held precision devices. An inertial measurement system captures translational and rotational vibrations in all six degrees of freedom across representative postures, instrument masses, and operators. The dataset is analyzed using unified spatial referencing, power spectral densities, statistical descriptors, and cross-spectral relationships to establish population-level vibration characteristics. The measurements indicate that operator-induced vibration is dominated by low-frequency physiological tremor, shows systematic dependencies on posture and mass, and exhibits minimal cross-axis coupling. Parametric spectral envelopes are fitted to the median spectra and expressed as normalized filters, enabling the synthesis of realistic vibration signals from appropriately scaled white noise. This approach provides simulation-ready disturbance inputs for control design, structural sensitivity analysis, and optomechatronic performance assessment. The proposed HAVC extend the vibration criterion concept from environmental excitation to human-instrument interaction, providing quantitative reference envelopes for the predictive design of hand-held precision systems.

1. Introduction

Vibration imposes fundamental limits on the performance of precision engineering systems. Gordon's seminal vibration-criterion (VC) curves [1] provide widely adopted limits for stationary laboratory and industrial environments by specifying allowable vibration velocity levels in one-third-octave bands. These curves serve as practical engineering tools for facility design, instrument siting, and vibration-isolation specification, ensuring stable operation of vibration-sensitive equipment such as optical benches, electron microscopes, metrology systems, and lithography tools [2,3].

In contrast, no analogous framework exists for hand-held precision instruments, despite their growing importance in scientific, medical, and industrial domains. In such systems, vibration generated by the human hand-arm system typically dominates over environmental vibration [4].

Standards for hand-transmitted vibration exposure are aimed at occupational health assessment and focus on high-vibration industrial tools such as grinders, hammers, and rotary power tools [5–7]. Their frequency weightings, exposure metrics, and amplitude ranges are inappropriate for the low-frequency, small-amplitude disturbances that limit the precision of handheld scientific instruments. Meanwhile, the biomedical tremor literature offers detailed analyses of pathological and physiological hand motion [8], but typically reports only dominant

tremor frequencies or scalar amplitude measures [9–11], and generally neglects rotational motion.

As a result, the design of hand-held precision systems faces a clear methodological gap. For instruments such as hand-held Fourier-transform infrared spectrometers, polarimeters, microscopes, interferometers, and optical alignment tools, whose measurement accuracy depends on internal or external spatial positioning, operator-induced vibration is often the primary disturbance source [12–14]. Depending on the operating principle, these vibrations may introduce (i) pointing errors, (ii) microphonic or acousto-electrical coupling in sensitive electronics, and (iii) relative displacement of internal measurement components [15–17]. Without quantitative descriptions of vibration for all six degrees of freedom (DoF), the predictive value of simulations, control-system design, and optomechanical analyses remains limited.

Furthermore, the vibratory behavior of the hand-arm system depends strongly on posture, instrument mass, and individual variability [18–20]. Capturing these effects requires spectral information across all translational and rotational degrees of freedom, expressed in a consistent spatial reference frame.

Beyond direct vibration characterization, recent work in engineering reliability and system-performance analysis increasingly relies on compact stochastic models that can be integrated into predictive simulation and robustness analysis [21,22]. In this context, disturbance

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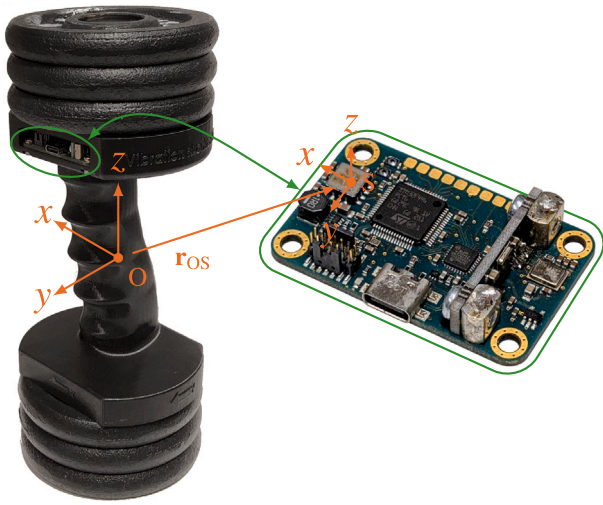


Fig. 1. Custom inertial measurement device, configured for 3.1 kg instrument mass (left). The electronic assembly is outlined in green in both views. The right image shows the sensing, readout, and interface components mounted in the top part, with the USB connection attached in the positive y -direction (not shown). The second gyroscope is mounted on an integrated riser card, allowing full 6-DoF sensing.

descriptions are useful not only for interpreting measurements, but also for controller evaluation and system-level performance prediction under realistic operating variability. The proposed HAVC follow this approach by expressing operator-induced vibration in a form suitable for model-based analyses of hand-held precision systems.

The contribution of this paper is the quantitative statistical assessment of physiological hand-arm vibrations and the introduction of generic hand-arm vibration criteria (HAVC) for hand-held precision instrumentation. A custom 6-DoF inertial measurement device is characterized for low uncertainty and used to record translational and rotational vibrations from 77 participants across three ergonomic postures and three representative instrument masses. Power spectral densities (PSDs), statistical envelopes, and cross-spectral relationships are analyzed to obtain a complete spatio-spectral description of operator-induced vibration. The resulting HAVC are formulated as parametric spectral-shaping filters that provide a compact, uncertainty-aware representation enabling the synthesis of realistic disturbance inputs for simulation, control testing, and instrumentation performance assessment. Rather than serving as pass/fail limits, the proposed HAVC provide population-level reference envelopes that reflect the dominant spectral characteristics of hand-held precision instrument operation and support improved prediction, evaluation, and design of hand-held precision instruments.

2. Measurement system and methodology

2.1. Inertial measurement device

A custom 6-DoF inertial measurement device (IMD) is developed to accurately record translational and rotational hand-arm vibrations. The central sensing core, as shown in Fig. 1, incorporates an Analog Devices ADXL355 tri-axial accelerometer and two ADXR290 dual-axis gyroscopes, arranged orthogonally to capture all three rotational degrees of freedom. The ADXL355 is configured for a full-scale range of $\pm 2g$, providing a noise density of $221 (\mu m/s^2)/\sqrt{Hz}$ and an integrated 20-bit delta-sigma ADC with an output filter bandwidth of 1 kHz. The output data rate is set to 4 kHz. Each ADXR290 provides a $\pm 100^\circ/s$ range, a noise density of $0.004 (^\circ/s)/\sqrt{Hz}$, and on-chip digitization with 16 bit resolution at an output data rate of 4.25 kHz with a bandwidth

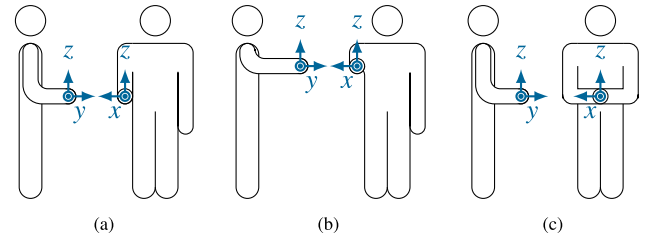


Fig. 2. Pictorial representation of the three investigated postures: (a) angled forearm posture (Posture 1), (b) extended upper-arm posture (Posture 2), and (c) two-hand angled posture (Posture 3).

of 480 Hz. The sensors are mounted on a four-layer printed circuit board (PCB) measuring $30 \times 45 \text{ mm}^2$, designed for precise orthogonality and minimal signal path length to reduce parasitic coupling and noise pickup.

All sensors interface digitally with an integrated ST Microelectronics STM32G474 microcontroller, which configures the sensors, synchronizes sampling, and aggregates data. To minimize sampling jitter and long-term frequency drift [23], the IMD uses a temperature-compensated crystal oscillator (TCXO) with a frequency tolerance of $\pm 100 \text{ ppb}$ and a phase noise of -80 dB at 1 Hz offset. The TCXO provides the master clock for both sampling and timestamping. Due to the unequal sampling frequencies for gyroscope and accelerometer channels, each sensor reading is augmented with a timestamp in oscillator clock cycles before it is transferred via USB 2.0 to a personal computer for storage and further analysis.

The PCB is mounted in a 3D-printed, grip-like handle fabricated from PET-G using fused deposition modeling (FDM), as shown in Fig. 1. The handle reproduces the geometry of typical hand-held precision instruments and provides a stable, reproducible grip interface. Appropriate mechanical tolerancing yields mounting tolerances below 1° (angular) and 0.2 mm (linear). The handle incorporates a magnetic mounting system that allows iron weights of 0.5 kg each to be attached symmetrically along the handle axis. This enables a minimum instrument mass of 0.1 kg (the handle's own weight) and additional loads in 1 kg increments, while preserving the handle's center of gravity (coinciding with the origin of the output coordinate system) and ergonomics. The magnetic attachment enables quick and repeatable reconfiguration between different masses, eliminating the need for mechanical fasteners and minimizing alignment errors.

Linear accelerations are measured along the orthogonal axes (x, y, z)_S of the local sensor reference frame. To enable reporting of results independent of the IMD's geometry, the output coordinate system (x, y, z)_O, centered in the participant's palm and illustrated in Fig. 1, is defined. In this reference frame, the y -axis is aligned longitudinally with the forearm, the x -axis is oriented laterally, and the z -axis is directed vertically upward. Rotational motion is characterized by the three Euler angles (α, β, γ), representing rotations about the respective x -, y -, and z -axis. While the measured angular velocities $\omega^T = [\dot{\alpha} \ \dot{\beta} \ \dot{\gamma}] = [\omega_\alpha \ \omega_\beta \ \omega_\gamma]$ are independent of the chosen coordinate system, the linear accelerations $\mathbf{a}^T = [a_x \ a_y \ a_z]$ are transformed to the output coordinate system via the relation

$$\mathbf{a}_O = \mathbf{a}_S + \boldsymbol{\alpha} \times \mathbf{r}_{OS} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}_{OS}). \quad (1)$$

Here, $\mathbf{r}_{OS}^T = [20 \text{ mm} \ 61.3 \text{ mm} \ 18.8 \text{ mm}]$ is the vector from the origin of coordinate system S to the origin of coordinate system O, and the angular acceleration vector $\boldsymbol{\alpha} = \dot{\boldsymbol{\omega}}$ is estimated using a 7-point cubic Savitzky-Golay filter.

This convention establishes a consistent spatial reference across participants and test conditions, thereby facilitating direct comparison of linear vibration amplitudes and angular displacements within a standardized coordinate framework.

Table 1

Moments of inertia in kg mm^2 for different system masses m in kg. Note that the resulting deviation moments L_{xy} , L_{xz} , and L_{yz} are independent of the system mass.

m	L_{xx}	L_{yy}	L_{zz}	L_{xy}	L_{xz}	L_{yz}
0.1	739.3	758.4	126.2	−0.2	−0.6	1.7
1.1	7901.2	7920.3	1590.8	−0.2	−0.6	1.7
3.1	30 572.5	30 591.7	4520.1	−0.2	−0.6	1.7

2.2. Experimental protocol

All measurements are conducted under controlled indoor conditions on a solid office floor to ensure constant vibration levels and provide reproducible measurement conditions across participants. Participants are instructed to maintain a relaxed yet stable upright stance during all trials, holding the instrument in their dominant hand and in a natural working position suitable for each posture.

Each participant carries out measurements in three standardized standing postures, shown in Fig. 2, selected to represent a wide range of ergonomic configurations commonly observed during precision instrument use [24]:

- *Angled forearm posture (P1)*: Standing upright with the elbow flexed to 90° , forearm angled forward, and the hand holding the instrument in a comfortable, natural position.
- *Extended upper-arm posture (P2)*: Standing upright with the arm fully extended horizontally at shoulder height, holding the instrument unsupported to simulate reach-oriented or alignment tasks.
- *Two-hand angled posture (P3)*: Standing upright with both forearms flexed to approximately 90° , holding the instrument bimanually in a symmetric configuration to mimic stable, high-precision handling.

For all three postures, participants are instructed to maintain a neutral wrist position and align the IMD's z-axis with the upward vertical direction. Grip force is neither measured nor controlled. Participants are instead instructed to hold the device naturally and stably to reflect typical hand-held precision instrument use.

To account for the varying inertial and ergonomic loads encountered across different hand-held systems, each posture is tested with three representative instrument masses:

- 0.1 kg: Simulating stylus- or pen-like devices, such as micropipettes, optical pointing devices, or fine mechanical styluses, typically used for high-dexterity operations with minimal load [25].
- 1.1 kg: Representing medium-weight portable sensors, such as hand-held FTIR spectrometers, polarimeters, or similar field-use analytical tools [24].
- 3.1 kg: Emulating compact analytical or optical instruments, such as portable microscopes or spectrographs, requiring bimanual stabilization due to their higher mass and inertia.

These three mass configurations are selected to span the practical weight range of precision hand-held instruments, encompassing both fine-manipulation and portable analytical applications. Table 1 lists the moments of inertia for all three used masses. The ratio of deviation moments to principal moments is below 1 %, effectively avoiding cross-axis coupling between the individual degrees of freedom.

Each of the nine measurements per participant (three masses times three postures) lasts 30 s, during which tri-axial accelerometer and gyroscope data are continuously recorded. A minimum rest interval of one minute is included between consecutive measurements to prevent muscular fatigue and maintain consistent performance throughout the experiments.

Table 2

Age and gender distribution for the participant cohort (47 males, 30 females; age range 18 to 65 years, mean 36.4 years, standard deviation 11.7 years).

Age group	18 to 25	25 to 35	35 to 45	45 to 55	55 to 65
Female	7	5	6	5	7
Male	5	17	11	9	5
Total	12	23	17	14	12

2.3. Participant cohort

It is well established that the magnitude and spectral characteristics of physiological tremor are influenced not only by individual-specific factors but also by various internal and external conditions, including muscle activation level, fatigue, limb posture, emotional state, temperature, and mechanical load [19,20,26]. To account for these variations, a group of 77 healthy adults with the age and gender distribution shown in Table 2 participated in the study. None of the participants reported any upper-limb or neuromuscular disorders. This cohort is considered representative of a healthy adult population and is therefore suitable for establishing generic vibration reference baselines.

2.4. Data processing

All recorded accelerometer and gyroscope signals are processed offline using a standardized pipeline to ensure uniform data quality and spectral comparability across participants and conditions. The data processing procedure comprises the following steps:

1. *Temporal resampling*: Raw sensor data are resampled to a uniform data rate of 4 kHz using linear interpolation to compensate for minor timing irregularities and to provide consistent temporal resolution for spectral analysis.
2. *Bandpass filtering*: The resampled signals are filtered using a zero-phase, 12th-order Butterworth bandpass filter with cutoff frequencies of 0.5 Hz to 550 Hz. This removes offset and drift errors that may be present at the sensor outputs. The upper bandwidth limit is chosen relatively high to avoid attenuation of potentially present vibration components. Zero-phase filtering (via forward-reverse application) is employed to prevent phase distortion and preserve the temporal characteristics of the vibration signals [27,28].
3. *Spectral analysis*: The one-sided power spectral density (PSD) is estimated over 0.5 Hz to 400 Hz with a frequency resolution of 0.5 Hz using Welch's averaged, modified periodogram method [29]. An FFT length of 8000 samples and a 50% segment overlap are used to achieve high frequency resolution while reducing variance between spectral estimates. Each segment is windowed with a Kaiser window ($\beta = 0.5$) to control spectral leakage [27]. The resulting PSDs (S_{xx} , S_{yy} , S_{zz} , S_{ax} , $S_{\beta\beta}$, $S_{\gamma\gamma}$) are expressed in units of acceleration or angular velocity squared per hertz ($(\text{m/s}^2)^2/\text{Hz}$ or $(^\circ/\text{s})^2/\text{Hz}$), respectively. Possible cross-axis coupling effects are assessed by computing the cross-spectral power density (CPSD) and normalizing it to the spectral correlation coefficient ρ .
4. *Statistical grouping and aggregation*: For each participant and test condition, PSD-derived vibration metrics are statistically grouped according to instrument mass (0.1 kg, 1.1 kg, and 3.1 kg) and posture (P1, P2, P3). For each group, time-domain metrics, as well as the median and corresponding 95 % confidence spectra are calculated to characterize the central tendency and variability of the vibration response across the cohort. Confidence intervals are derived using the bootstrap-t method [30].

This processing pipeline yields consistent, high-resolution spectral representations of both translational and rotational hand–arm vibration components, providing a reliable basis for subsequent analysis.

2.5. Measurement uncertainty

Sensor-intrinsic noise is characterized using static measurements on a pneumatic vibration-isolation table, yielding noise floors of $17.7 \cdot 10^{-9} (\text{m/s}^2)^2/\text{Hz}$ for linear acceleration and $3.16 \cdot 10^{-6} (^\circ/\text{s})^2/\text{Hz}$ for angular velocity. Allan variance analysis [31] confirms stable drift behavior over the 30 s measurement window, and calibration measurements on a precision shaker setup showed amplitude linearity within ± 1 dB and cross-axis sensitivities below 2% across the relevant bandwidth up to 400 Hz. These results demonstrate that the overall measurement uncertainty is sufficiently low for the following vibration spectra to reliably reflect true operator-induced motion.

2.6. Data privacy and ethics

All participant data are fully anonymized before analysis. A randomized ID is assigned to each subject, and no personally identifiable information is stored with the measurement data. All participants provided written informed consent in accordance with the principles outlined in the Declaration of Helsinki [32].

3. Experimental results

3.1. Vibration magnitudes

Fig. 3 shows the measured vibration magnitudes across all test conditions, presented as boxplots of RMS linear accelerations and angular velocities for the three instrument masses (0.1 kg, 1.1 kg, and 3.1 kg) and the three postures. Each plot shows the distributions of translational components (a_x , a_y , a_z) and rotational components (ω_α , ω_β , ω_γ).

Across all configurations, median vibration levels are consistently in the range of 0.1 m/s^2 to 1 m/s^2 for linear acceleration and $0.1^\circ/\text{s}$ to $1^\circ/\text{s}$ for angular velocity. These results indicate a stable vibration environment representative of controlled precision-handling conditions.

However, broad variability is observed across participants, with minimum RMS values of (0.0230, 0.0227, 0.0254, 0.0226, 0.0250, 0.0259) and maximum RMS values of (0.8523, 0.9827, 1.1338, 1.0439, 2.4872, 1.0348) for the six measured components (a_x , a_y , a_z , ω_α , ω_β , ω_γ). This spread reflects the combined influence of tool mass, posture, and individual handling variability.

A slight dependence on instrument mass is evident. The heavier configurations tended to produce larger inter-quartile ranges, indicating increased influence of participant physique. Postural influences are moderate but discernible. The extended-arm posture (Posture 2) results in slightly elevated vibration magnitudes compared to the angled forearm (Posture 1) and two-hand angled (Posture 3) postures. This increase may be attributed to reduced mechanical support and higher muscular effort during unsupported arm extension, which may lead to increased involuntary motion components.

Overall, the box plots show that vibration amplitudes remain within a consistently low magnitude range across all tested configurations, confirming the consistency of the measurement procedure and providing a representative quantitative description of the hand–arm vibration characteristics during precision instrument handling.

3.2. Spectral characteristics

Fig. 4 presents the spectral power density (PSD) results for all investigated postures and instrument masses. Each subfigure displays the translational (S_{xx} , S_{yy} , S_{zz}) and rotational ($S_{\alpha\alpha}$, $S_{\beta\beta}$, $S_{\gamma\gamma}$) components for the three mass configurations (0.1 kg, 1.1 kg, and 3.1 kg).

Across all measurement conditions, the spectra exhibit a characteristic low-pass behavior, with most vibration energy concentrated at frequencies below approximately 50 Hz. This trend reflects the natural

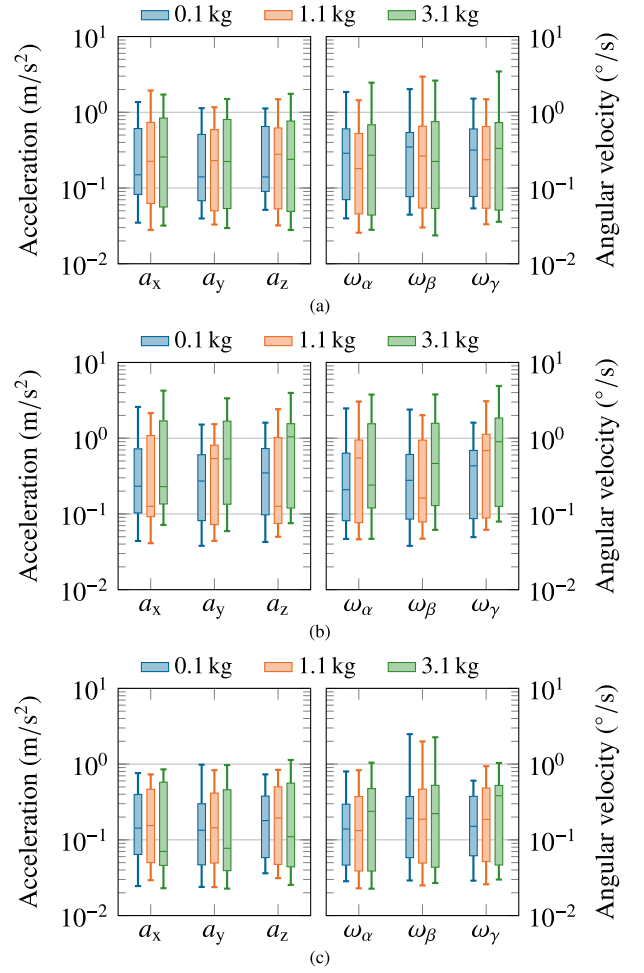


Fig. 3. Box plots of RMS linear accelerations and angular velocities for different mass configurations in (a) Posture 1, (b) Posture 2, and (c) Posture 3. Each plot shows the distributions of translational (x , y , z) and rotational (α , β , γ) components under loads of 0.1 kg, 1.1 kg, and 3.1 kg.

damping characteristics of the hand–arm system, in which higher-frequency motion components are strongly attenuated by muscular and soft-tissue compliance as well as voluntary control [33].

A clear dependence on instrument mass is observed. The lightest configuration (0.1 kg) produces the lowest overall PSD values while exhibiting the highest cutoff frequency, indicating that lightweight instruments allow for faster motion components due to their low inertia. In contrast, the heaviest configuration (3.1 kg) shows the highest PSD values at low frequencies and a lower cutoff frequency, consistent with the dominance of inertial effects and reduced bandwidth of voluntary motion when handling heavier instruments.

Additionally, the 3.1 kg configuration displays a distinct secondary spectral peak around 50 Hz to 70 Hz, observable across several postures and axes. This feature is attributed to muscle tremor, a physiological phenomenon originating from involuntary oscillations in muscle tension during sustained static effort [10]. The appearance of this component suggests a measurable neuromuscular contribution to the vibration signal under conditions of increased load. Because this feature is most pronounced under heavy-load conditions and is not consistently present across all cases, the generic envelope introduced below is formulated to capture the dominant broadband trend rather than local spectral features.

To assess potential cross-axis coupling effects, the CPSD is computed for all translational and rotational channel pairs. No significant

median correlations are found across participants or conditions, where typical spectral correlation coefficients of $|\rho| < 0.2$ indicate that vibrations along different axes are largely uncorrelated and directionally independent.

Overall, the spectral results demonstrate that vibration during precision handling is dominated by low-frequency components associated with voluntary control, with spectral bandwidth and secondary tremor contributions influenced primarily by instrument mass. The weak inter-axis coupling confirms that hand–arm vibration in precision tasks is largely axis-specific and mechanically decoupled.

3.3. Generic vibration envelopes

To derive compact models that capture the characteristic spectral behavior of the measured hand–arm vibrations, generic vibration envelopes are constructed by fitting parametric low-pass filters to the median PSDs obtained from all measurement conditions. For each translational and rotational component, the transfer function

$$G(s) = g_0 \left(\frac{1}{1 + s/(2\pi f_c)} \right)^k \quad (2)$$

is fitted, where f_c denotes the cutoff frequency, k the filter order, and g_0 a scaling factor. To ensure comparability across axes, the gain g_0 is determined such that for white input spectra, the filter's output and input power are equal. Specifically, the value

$$P = \int_{-\infty}^{\infty} |G(j2\pi f)|^2 df = g_0^2 f_c \sqrt{\pi} \frac{\Gamma(k - 1/2)}{\Gamma(k)}, \quad (3)$$

where $\Gamma(\cdot)$ denotes the gamma function [34], is required to satisfy the normalization condition $P = 1$ for each axis. This ensures that the fitted filters describe only the shape of the spectral envelope, independent of the absolute vibration variance. The filter order k is chosen as the minimum order, so that the coefficient of determination R^2 exceeds 0.95.

The fitted parameter sets for all axes are summarized in Table 3, which additionally lists the corresponding R^2 values. These are found to be satisfactory across all axes, indicating that the proposed filter structure effectively captures the overall spectral shape. The filter order is found as $k = 2$ for the linear degrees of freedom (x , y , z) and $k = 6$ for the rotational degrees of freedom (α , β , γ).

Although a secondary peak at 50 Hz to 70 Hz is observed under 3.1 kg loading in some postures and degrees of freedom, a simple low-pass filter is used to capture the dominant broadband trend. This allows a single parametric transfer function to be applied across all conditions while preserving the main spectral characteristics. Applications that require a more accurate representation of the secondary peak may include an additional parallel filter term.

Using these normalized envelopes, the measured vibration signals can be reproduced synthetically. White Gaussian noise with unit PSD is first passed through a gain block that sets the desired vibration level (e.g., median variance or 95 % confidence variance). The scaled noise is then fed into the fitted filter $G(s)$, producing time-domain signals whose spectral characteristics match the corresponding median vibration envelopes. When applying the filter transfer functions to 95 % confidence variances, R^2 values stay above 0.93, supporting the chosen fit.

Because an independent noise source drives each axis, the resulting synthetic vibration signals are uncorrelated across axes. This is consistent with the empirical findings reported in the previous subsection, where CPSD analysis showed only very weak cross-axis coupling (typically $|\rho| < 0.2$).

3.4. Application of HAVC

The HAVC can be used directly in simulation, control development, and optomechanical analysis of hand-held precision instruments. Their

implementation requires only three elements: (i) a white-noise source, (ii) an amplitude scaling factor, and (iii) the axis-specific spectral-shaping filter $G(s)$. The practical workflow shown in Fig. 5 can be implemented in any suitable simulation environment:

1. *Posture and mass selection:* From the nine reported posture-mass configurations, select the one that best represents the simulation scenario and obtain the corresponding filter parameters from Table 3.
2. *RMS level selection:* Decide whether the simulation should reflect a typical or a conservative scenario and choose the corresponding RMS level σ from Figs. 3(a)–3(c). For realistic operating conditions, the median value can be selected, whereas an upper-percentile value provides a more conservative estimate.
3. *Noise generation:* For all six degrees of freedom, generate independent white Gaussian noise that acts as the model input. Unit PSD ensures consistent normalization across axes.
4. *Magnitude scaling:* Scale the generated noise signals with the standard deviation σ to set the level of the resulting vibration.
5. *Spectral shaping:* For each axis, pass the scaled noise signals through the HAVC filter $G(s)$ with the previously selected transfer function parameters (k , g_0 , f_c).
6. *Noise injection:* Inject the synthesized signals into the simulation model as an external input.

This procedure yields synthetic time-domain vibration signals whose spectral content matches the measured human-induced vibration. A representative example for the x -axis in Posture 1 with a 1.1 kg load is shown in Fig. 6. The synthetic PSD reproduces the measured median spectrum, confirming that the filter-based synthesis accurately captures the dominant spectral characteristics.

Because all HAVC filters are normalized to unit power transfer, the applied gain σ fully determines the RMS value of the generated vibration. This makes it easy to adjust the disturbance level without altering the filter structure. A practical advantage of the parametric representation is that it enables straightforward computation of derived motion quantities. Since $G(s)$ is an analytic continuous-time transfer function, linear velocity, displacement, angular displacement, and angular acceleration can be obtained through simple integration or differentiation inside the simulation environment. This avoids numerical drift and filtering artifacts that occur when deriving these signals directly from measured data.

4. Discussion

This work extends Gordon's vibration-criterion concept from stationary environments to the dynamic context of hand-held instruments. While Gordon's curves describe maximum allowable environmental vibration levels in one-third-octave velocity bands, the present study instead characterizes operator-induced vibration using full PSDs of linear acceleration and angular velocity. The obtained envelopes should therefore be interpreted as reference spectra rather than acceptance criteria.

Several trends consistently emerge across the 6-DoF dataset. First, instrument mass and posture have a meaningful influence on vibration magnitude. Heavier tools and extended-arm postures increase vibration levels, consistent with higher muscular activation, reduced biomechanical damping, and the need to sustain static joint torques. Lighter tools lead to higher vibration bandwidth, whereas heavier instruments shift spectral energy toward lower frequencies, revealing a distinct secondary spectral peak associated with muscle tremor. Second, the vibration spectra are remarkably stable in shape across participants and conditions. All spectra exhibit a characteristic low-pass form dominated by physiological tremor within approximately 8 Hz to 14 Hz, with diminishing content above 50 Hz. The rotational spectra mirror this low-frequency dominance, although with higher corner

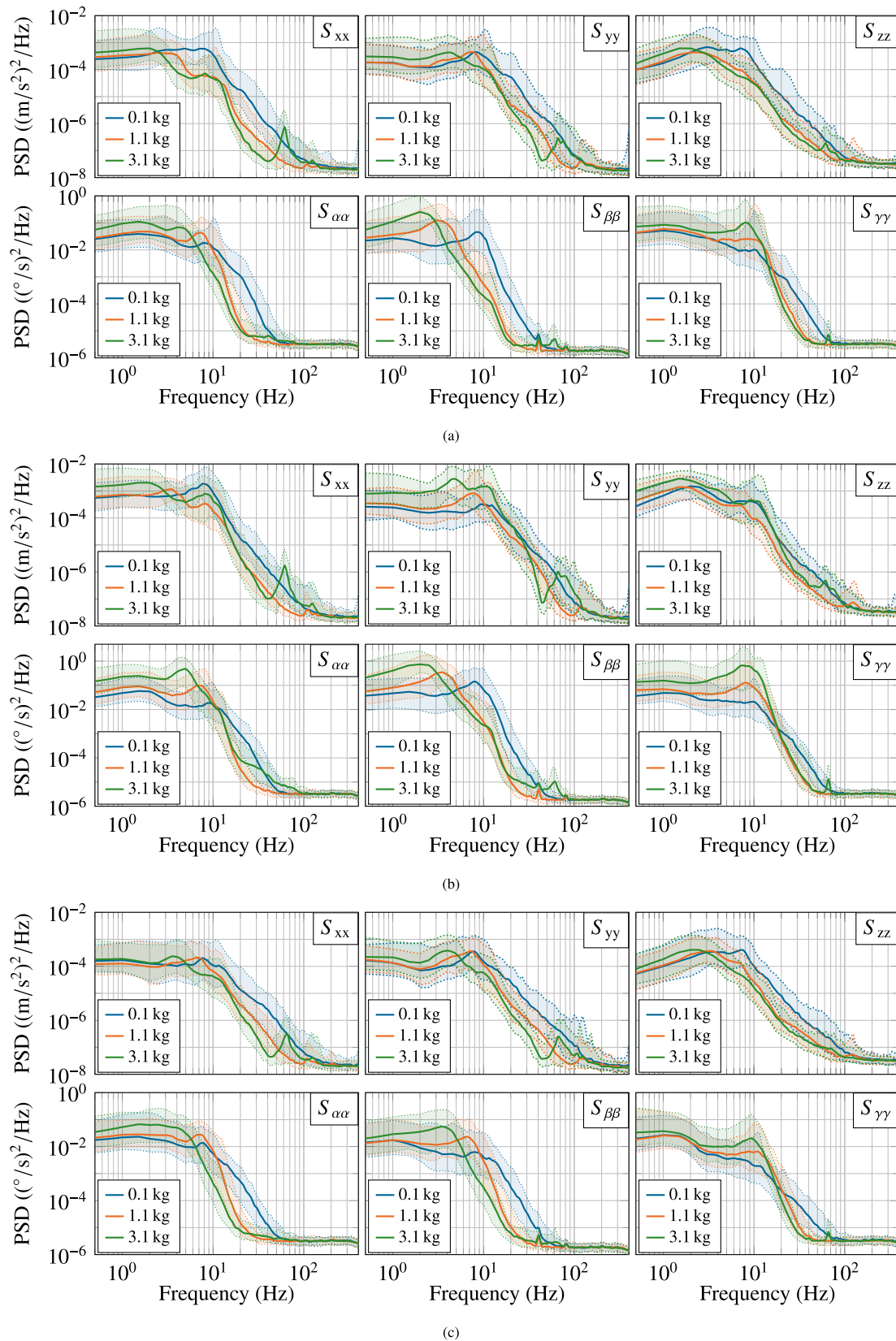


Fig. 4. PSD results for all investigated postures. Each subfigure presents the translational components (S_{xx} , S_{yy} , S_{zz}) and the rotational components ($S_{\alpha\alpha}$, $S_{\beta\beta}$, $S_{\gamma\gamma}$) for different mass configurations (0.1 kg, 1.1 kg, and 3.1 kg). Panels (a)–(c) correspond to postures P1, P2, and P3, respectively. The solid line indicates the median value, and the shaded area represents the 95% confidence interval.

Table 3

Fitted filter parameters for the generic vibration envelopes, including cutoff frequency f_c and normalized gain g_0 . The table also reports the coefficient of determination R^2 for each axis, demonstrating satisfactory agreement between the fitted transfer functions and the measured median PSDs.

		Posture 1			Posture 2			Posture 3		
		0.1 kg	1.1 kg	3.1 kg	0.1 kg	1.1 kg	3.1 kg	0.1 kg	1.1 kg	3.1 kg
x	g_0	0.26	0.34	0.38	0.27	0.34	0.34	0.21	0.27	0.30
	f_c (Hz)	9.53	5.46	4.30	8.64	5.52	5.38	14.12	8.95	6.91
	R^2	0.98	0.99	0.95	0.97	0.98	0.95	0.99	0.98	0.97
y	g_0	0.21	0.26	0.30	0.20	0.25	0.27	0.20	0.26	0.31
	f_c (Hz)	14.79	9.37	7.17	15.21	10.01	8.88	15.92	9.30	6.72
	R^2	0.98	0.97	0.97	0.98	0.97	0.95	0.98	0.97	0.97
z	g_0	0.25	0.31	0.35	0.28	0.34	0.33	0.24	0.30	0.34
	f_c (Hz)	9.89	6.48	5.34	8.23	5.38	5.68	11.05	7.01	5.58
	R^2	0.97	0.99	0.99	0.98	0.99	0.98	0.96	0.97	0.98
α	g_0	0.24	0.31	0.36	0.24	0.31	0.31	0.24	0.31	0.38
	f_c (Hz)	22.70	13.19	9.97	22.90	13.57	13.11	23.10	13.31	9.19
	R^2	0.99	0.98	0.99	0.99	0.98	0.97	0.99	0.98	0.99
β	g_0	0.26	0.37	0.43	0.26	0.36	0.39	0.23	0.31	0.37
	f_c (Hz)	18.50	9.58	7.06	18.90	10.21	8.48	23.80	13.39	9.50
	R^2	0.98	0.99	0.98	0.97	0.99	0.97	0.99	0.98	0.98
γ	g_0	0.22	0.27	0.29	0.22	0.26	0.27	0.23	0.25	0.27
	f_c (Hz)	26.00	18.10	15.57	27.20	19.00	17.20	25.00	20.60	17.40
	R^2	0.98	0.99	0.98	0.99	0.98	0.96	0.95	0.98	0.98

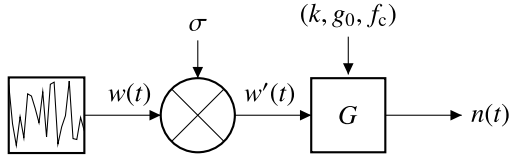


Fig. 5. Workflow for generating a synthetic vibration signal for a single degree of freedom. White Gaussian noise $w(t)$ is scaled to the desired level $w'(t)$ and shaped by the corresponding HAVC filter $G(s)$ to generate the desired vibration signal $n(t)$. The procedure is applied identically and independently to all six degrees of freedom.

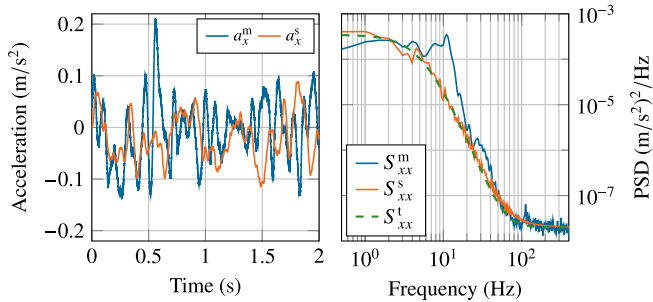


Fig. 6. Comparison between measured and simulated vibration signals. The left panel shows time-domain acceleration traces in the x -direction for the measured signal a_x^m and the synthesized signal a_x^s . The right panel presents their corresponding PSDs S_{xx}^m and S_{xx}^s , together with the target spectrum $S_{xx}^t = \sigma^2 |G(j2\pi f)|^2$.

frequencies up to approximately 25 Hz, highlighting the importance of rotational motion for accurate hand-tool dynamics modeling. Third, cross-spectral analysis shows only weak coupling between translational and rotational components. The low median spectral correlation coefficients (typically $|\rho| < 0.2$) indicate that tremor-induced vibrations along different axes arise primarily from independent neuromuscular mechanisms rather than from strongly coupled body dynamics. This validates modeling approaches that treat the six degrees of freedom as statistically independent excitation sources. Finally, expressing operator-induced vibration as parametric PSD envelopes, rather than

exposure-weighted metrics or isolated spectral peaks, allows the proposed HAVC to serve as simulation-ready disturbance models. These envelopes are therefore directly compatible with control-system simulations, finite-element structural models, optomechanical sensitivity analyses, and disturbance-rejection studies. More broadly, this representation is compatible with model-based robustness analysis and performance evaluation workflows, where compact stochastic disturbance models are needed for predictive evaluation of system robustness under realistic operating conditions [35,36].

Although the proposed HAVC provide a full 6-DoF characterization of operator-induced vibration, several limitations must be acknowledged. First, all measurements are acquired in static postures and on a stable indoor floor. Dynamic tasks, varying environmental conditions, and motion-intensive workflows are not considered. Second, only healthy adult participants are included, and factors such as fatigue, grip force variability, and task-specific cognitive load are not explicitly controlled or quantified. Since grip force is a proxy for muscle activation and may influence tremor amplitude and spectral shape, certain mass-dependent changes, especially the heavy-load peak, may reflect both inertia and neuromuscular activation [37,38]. Third, a single instrument geometry and handle design are used, which may influence biomechanical coupling and therefore limit generalization to tools with substantially different form factors. Finally, while sensor calibration and uncertainty are carefully addressed, the HAVC represent population-level statistical envelopes, and individual users may exhibit vibration characteristics outside the reported ranges. These constraints should be considered when applying the proposed envelopes to specific measurement scenarios or bespoke instrument designs.

The HAVC should therefore be regarded as population-level reference envelopes, conceptually analogous to Gordon's generic curves: not prescriptive limits, but broadly applicable baselines for realistic disturbance modeling in hand-held precision instrumentation.

5. Conclusion

This paper presents a systematic 6-DoF characterization of hand-arm vibration using a custom inertial measurement device and a cohort of 77 adults across nine posture-mass combinations. Unlike prior biomedical tremor studies, the dataset captures both translational and rotational spectral content with high bandwidth and consistent spatial referencing. The results reveal stable low-frequency spectral characteristics dominated by physiological tremor, weak cross-axis coupling, and systematic dependencies on tool mass and ergonomic posture.

Based on these findings, the hand–arm vibration criteria (HAVC) are introduced: population-based vibration reference envelopes that extend Gordon's vibration-criterion philosophy to hand-held precision instruments; they are intended for disturbance modeling and synthesis rather than as prescriptive pass/fail limits. The envelopes are provided as normalized, parametric PSD models that can be combined with arbitrary vibration amplitudes to generate synthetic, simulation-ready disturbance inputs. This simple model structure and the decoupling of spectral shape and magnitude facilitate straightforward integration into mechanical, electrical, or control systems simulation environments, therefore supporting the predictive design and analysis of hand-held metrological systems, such as FTIR spectrometers, polarimeters, microscopes, and other hand-held optomechatronic devices.

Although no single dataset can capture the full variability of human tremor across all individuals and use cases, the HAVC offer a practical and generalizable framework for representing operator-induced vibration in engineering analyses. The proposed envelopes thereby connect biomedical tremor characterization, occupational vibration standards, and the requirements of modern hand-held precision instrumentation.

In summary, the HAVC provide a compact and flexible tool for injecting realistic 6-DoF human-induced vibration into simulation models.

CRedit authorship contribution statement

Johannes Wiesböck: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Georg Schitter:** Writing – review & editing, Supervision, Resources, Conceptualization. **Ernst Csencsics:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Johannes Wiesböck reports financial support was provided by Austrian Research Promotion Agency (FFG). Johannes Wiesböck reports financial support was provided by Grabner Instruments Messtechnik GmbH. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors gratefully acknowledge financial support from Grabner Instruments Messtechnik GmbH and the Austrian Research Promotion Agency (FFG; grant number FO999891715). The authors also thank all participants for their contributions to the data collection and analysis.

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